

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV

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Mock Interview

1. What is Eigenfaces?

Answer: Eigenfaces is a PCA-based face recognition technique that represents faces as a linear combination of a small set of eigenvectors (the "eigenfaces") derived from the covariance matrix of a training set of face images. Each face is compactly represented by its projection weights onto these eigenvectors (a point in "face space"), and recognition is performed by comparing a probe face's weight vector to those of known faces, typically using Euclidean distance.

2. Define PCA in image processing.

Answer: Principal Component Analysis (PCA) is a dimensionality-reduction technique that finds the orthogonal directions (principal components) along which data varies the most. In image processing, images are treated as high-dimensional vectors; PCA computes the covariance matrix of a training set and projects each image onto the top-k eigenvectors of that matrix, retaining most of the variance in far fewer dimensions. This underlies techniques such as Eigenfaces, and is also used for image compression, denoising, and general feature extraction.

3. What is contour-based shape representation?

Answer: Contour-based shape representation describes an object using only its boundary (outline) rather than its interior pixels — for example, chain codes, boundary polygons, or Fourier descriptors fitted to the edge points. It is compact and works well when boundaries are clean and well-defined, but is more sensitive to noise, gaps, or occlusion along the boundary since it discards all interior information.

4. What is region-based shape representation?

Answer: Region-based shape representation describes an object using all the pixels within its area, not just the boundary — for example, geometric moments (including Hu moments), area, and pixel-density descriptors. It is more robust than contour-based methods to noisy or broken boundaries and can naturally handle shapes with holes or disconnected regions, at the cost of typically being more computationally expensive to compute over the full region.

5. How does an active contour (snake) work?

Answer: An active contour, or "snake," is a deformable curve that is initialized near an object's boundary and then iteratively evolves to minimize an energy functional. The energy has an internal term (encouraging smoothness and continuity of the curve) and an external term (usually derived from image gradients, pulling the curve toward strong edges). The curve is updated step by step — e.g., via gradient descent — until it settles into a local energy minimum that tightly follows the object's true boundary.

6. How does a level set method work?

Answer: A level set method represents a contour implicitly as the zero level set of a higher-dimensional function (commonly a signed distance function) rather than as an explicit parametric curve. The contour is evolved by updating this embedding function according to a partial differential equation driven by curvature and image-based speed terms. Because

the boundary is only implicitly defined, level sets naturally handle topological changes — such as a single contour splitting into two, or two contours merging — which explicit snakes handle poorly.

7. What is the Hough Transform?

Answer: The Hough Transform is a technique for detecting parametric shapes, most commonly straight lines (and also circles or other curves), in an image. Each edge point "votes" for every possible shape parameter it could belong to (e.g., every (ρ, θ) line passing through it), and these votes are accumulated in a parameter-space array. Peaks in this accumulator correspond to shapes that are actually present in the image, making the method robust to noise and partial occlusion since it aggregates evidence across many points rather than relying on any single one.

8. How are Fourier descriptors used in shape analysis?

Answer: A closed shape's boundary is traced and represented as a sequence of complex numbers $(x + iy)$ sampled along the contour; applying the Discrete Fourier Transform to this sequence yields a set of Fourier coefficients. After normalizing for starting point, scale, rotation, and translation, these coefficients form a compact, invariant descriptor of the shape, and shapes can then be compared or classified by comparing their Fourier descriptor vectors rather than raw boundary points.

9. How are wavelet descriptors used in shape analysis?

Answer: Wavelet descriptors serve a similar purpose to Fourier descriptors but use a wavelet transform of the shape boundary (or image) instead of a Fourier transform. Because wavelets provide multi-resolution analysis — capturing both frequency and spatial/local information — wavelet descriptors can represent coarse overall shape as well as fine local details at multiple scales simultaneously, which is useful when a shape's identity depends on both its overall form and small local features.

10. What is a particle filter?

Answer: A particle filter is a sequential Monte Carlo technique for estimating the state of a system (e.g., an object's position and velocity) under non-linear, non-Gaussian dynamics, commonly used for visual tracking. The posterior distribution over the object's state is approximated by a set of weighted random samples ("particles"); at each time step the particles are propagated forward using a motion model, re-weighted according to how well they match the current observation (e.g., appearance likelihood), and then resampled to concentrate particles in high-probability regions. This makes particle filters effective for tracking through occlusion, clutter, and multi-modal ambiguity that simpler trackers (like a single Kalman filter) struggle with.

11. How is foreground–background separation done in surveillance systems?

Answer: Foreground-background separation, or background subtraction, segments moving objects from a largely static background in video. A common classical approach models each pixel's intensity over time with a statistical model — most notably a Gaussian Mixture Model (GMM) per pixel — which adapts to gradual scene changes (like lighting drift); a pixel is flagged as foreground when its current value deviates significantly from the learned background model. More recent systems replace or augment this with deep learning-based background subtraction or semantic segmentation for greater robustness to shadows, dynamic backgrounds, and camera motion.

12. How do multi-camera systems track pedestrians?

Answer: Multi-camera pedestrian tracking typically works in three stages: (1) per-camera detection, using a person detector to produce bounding boxes in each frame; (2) single-camera tracking, associating detections across consecutive frames using motion prediction (e.g., a Kalman filter) plus appearance cues to form per-camera tracklets; and (3) cross-camera re-identification, matching Re-ID feature embeddings between tracklets from different cameras — often combined with knowledge of camera topology and plausible transition times — to maintain a single consistent identity as a person moves between camera views.

13. What is a Convolutional Neural Network (CNN)?

Answer: A Convolutional Neural Network is a deep learning architecture built from layers of learnable convolutional filters that slide across the input, exploiting spatial locality and weight sharing. Successive layers of convolution, non-linearity, and pooling build up a hierarchy of features — from simple edges and textures in early layers to parts and whole objects in deeper layers — making CNNs the long-standing standard architecture for image classification, object detection, and segmentation.

14. How does a Vision Transformer (ViT) work?

Answer: A Vision Transformer applies the Transformer architecture, originally developed for NLP, to images. The image is split into fixed-size patches, each patch is linearly embedded into a token vector and combined with a positional embedding, and the resulting sequence of tokens is processed through stacked self-attention layers. Because self-attention relates every token to every other token from the very first layer, a ViT captures global context immediately, in contrast to a CNN's local receptive fields, which only build up global context gradually with depth.

15. What is a Generative Adversarial Network (GAN)?

Answer: A Generative Adversarial Network consists of two neural networks — a generator and a discriminator — trained together in an adversarial, minimax game. The generator takes random noise as input and tries to produce synthetic samples realistic enough to fool the discriminator, while the discriminator tries to correctly distinguish real training samples from the generator's fakes. As training proceeds, this competition drives the generator to produce increasingly realistic outputs, and GANs are widely used for image synthesis, data augmentation, image-to-image translation, and super-resolution.

16. How is a Support Vector Machine (SVM) used for image classification?

Answer: A Support Vector Machine finds the hyperplane that separates classes with the maximum margin in feature space, optionally using a kernel (e.g., RBF) to handle classes that are not linearly separable in the original space. For image classification, an SVM is typically applied on top of extracted features — classical descriptors such as HOG or SIFT, or embeddings from a pretrained CNN — rather than directly on raw pixels, and tends to perform particularly well on smaller datasets with well-engineered or well-pretrained features.

17. How are Hidden Markov Models (HMM) applied in activity recognition?

Answer: A Hidden Markov Model models a sequence of observed features (e.g., pose or motion features extracted per video frame) as being generated by an underlying, unobserved sequence of hidden states (e.g., phases of an activity), governed by state transition probabilities and per-state observation probability distributions. One HMM is typically trained per activity class; recognizing an observed sequence then involves computing which model's HMM best explains it (e.g., via the forward algorithm), or decoding the most likely underlying state sequence with the Viterbi algorithm, making HMMs well suited to activities with a natural temporal/sequential structure.

18. What is a Gaussian Mixture Model (GMM)?

Answer: A Gaussian Mixture Model represents a probability distribution as a weighted sum of several Gaussian components, each with its own mean, covariance, and mixing weight. Parameters are typically estimated with the Expectation-Maximization (EM) algorithm. In computer vision, GMMs are widely used for per-pixel background modeling in surveillance (each pixel's intensity history modeled as a mixture of Gaussians), as well as for image segmentation and general soft/probabilistic clustering, where a sample can have partial membership across multiple components rather than a single hard label.

19. How is graph-based learning applied to 3D shape correspondence?

Answer: Graph-based learning represents a 3D mesh as a graph of vertices and edges and establishes correspondences between points on different meshes using the mesh's intrinsic (graph/spectral) structure rather than requiring a fixed, consistent vertex ordering across meshes. Approaches include spectral/functional maps built from the Laplace-Beltrami eigenbasis, and graph neural networks that learn correspondence-relevant features via message passing over the mesh graph. This makes graph-based methods considerably more robust than a fixed-topology model (such as a linear PCA shape prior) to meshes with differing topology, vertex density, and non-rigid deformation.

20. How is road and lane detection performed in autonomous driving?

Answer: A typical lane detection pipeline preprocesses each video frame (grayscale conversion, Gaussian blur, sometimes shadow-robust contrast normalization), applies edge detection (commonly Canny), restricts the search to a region of interest covering the road, and then either fits lane geometry classically (Hough Transform to find line segments, followed by a polynomial fit to allow for curves) or uses a deep learning semantic segmentation model trained to classify lane pixels directly. Robust systems additionally apply temporal smoothing across frames and fall back on recent detections during brief partial occlusion (e.g., another vehicle temporarily covering a lane marking), so the detected lane boundary remains stable frame to frame.